TJWALKER +
ASSOCIATES INC

January 1, 2017

Subject:
TJWalker + Associates Inc. - Web Handling Consulting Services

Thank you for your interest in my web handling consulting services.

The attached pages provide the information about my background, consulting, and rates.

Please find attached:

- My Web Handling and Consulting Philosophies
- Web Handling Sub-Technology Areas
- Consulting Rates and Example Estimate
- Brief Resume
- Publications / Presentations
- Outline of Web Handling Fundamentals Seminar
  (in separate file, if requested)

Please call or email if you have questions.

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TJWalker + Associates Inc. Web Handling Consulting Services

Web Handling and Consulting Philosophies

I have worked with many different web processes from wide, high speed operations for commodity products to small, precision manufacturing for battery, solar, flexible electronics, packaging, printing, and biomedical applications. Though my specialty is thin film web handling, I find the engineering principles of web handling apply to any web product, including papers, films, foils, nonwovens, textiles, and any combination of these in coated or laminated form.

Since 2001, I have worked with over 180 companies to improve their web handling processes or train their technical staffs. I keep my client information confidential, but have several past clients that will gladly serve as references upon request.

The goal of the web handling engineer is to transport and store a web product while minimizing waste. Winding, unwinding, and rewinding are inherent waste generators. Process integration eliminates winding waste. There is a practical limit where width, speed, complexity, or other incompatibilities prevent integration, but otherwise web handling should not limit integration.

Web handling engineers must also work with process and product challenges. Webs are inherently imperfect, with both thickness and length variations. Good web handling processes are designed with an understanding of the imperfect web.

The best place to start designing a web process is the product development stage. I believe in *design for manufacturing*; knowledge of web handling and roll-to-roll processes will pay dividends in production costs and yields.

I am also a strong advocate for assessing profitability. I believe teams responsible for technology planning and prioritization should also understand revenue, costs, and market share. Just as manufacturing knowledge helps product design; profit strategy knowledge helps engineering – *design for profitability*.

With this in mind, I want to work with clients where my services will have a high potential to have a significant and timely return on investment.

Areas where my services may prove valuable:

- Process optimization for productivity or waste
- Technical training for engineers, operators, and maintenance
- Product and process development
- Specification development and vendor selection
- Equipment checkout, calibration, and installation
- Documentation for process procedures, training, or patenting

Web Handling Sub-Technology Areas

I divide web handling technology into the following areas:

- Web properties / behavior
- Tension and speed control, including machine direction registration
- Roller performance and alignment
- Nipped roller systems
- Web-to-roller traction / friction / adhesion / lubrication
- Guiding (lateral registration)
- Web wrinkling and spreading
- Air flotation (replacing rollers with air cushions)
- Laminating
- Winding, Winders, and Wound Rolls
- Slitting

The goal of web handling is to transport and position the web through integrated processes. Though this doesn't directly add value, poor web handling can easily create waste.

The top areas of web handling related waste are:

- ✓ Tension variations, including:
  too high or too low,
  variations over time,
  variations through rollers,
  variations across web width, and
  variations in slit strands after slitting.
- ✓ Web non-uniformity (bagginess, camber, and curl or non-lay-flat)
- ✓ Web-roller slip (including abrasion / scratching, loss of control)
- ✓ Air lubrication on rollers and air entrainment in winding rolls
- ✓ Lateral shifting in handling, misalignment, or registration error
- ✓ Buckling in span between rollers (troughing, buckling while on a roller, (wrinkling)
- ✓ Roll cinching and telescoping (both at winding and unwinding)
- ✓ Crossweb buckles or wrinkles in rolls
- ✓ Machine direction ridges in rolls (tin-canning)
- ✓ Impressions and contamination in rolls or handling
- ✓ Sidewall misalignment in rolls (other than cinching / telescoping)
- ✓ Winding defects associated with crossweb thickness variations (bagginess and skew)
- ✓ Web breaks (especially for paper products)
- ✓ Non-uniform nip pressure (leading to wrinkles, bubbles, or curl)

Timothy James Walker

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Areas of Expertise

1. Web handling process technology (development, optimization, and education)
2. Process development and manufacturing problem solving
3. Technology management and transfer

Experience

TJWalker + Associates Inc., President 1999-present
St Paul, MN (2004-present), Decatur, GA (2002-2004),
St. Louis, MO (1999-2002)

- Providing web handling process expertise to the paper, film, and foil converting industry.
- Presenting web handling process technology seminars at technical conferences and direct to client engineering groups.
- Contributing Editor for Paper, Film, and Foil Converter magazine (since Feb 2002).
- AIMCAL (Assn. of Industrial Metalizers, Coaters, & Laminators) Technical Panel Board member (2002-2012).
- CEMA (Converting Equipment Manufacturers Association) Board of Trustees (2007-2012)
- Project management, as needed, to assist clients with implementing recommendations.

3M Company, Process development specialist / supervisor 1983-1999
St. Paul, MN

- Led corporate web handling process technology staff group, providing process research, production problem solving, equipment design reviews, and internal education program.
- Served as corporate representative to Oklahoma State University Web Handling Research Center Industrial Advisory Board.
- Led corporate-wide technology strategic planning and transfer program.

Education

- MS Management of Technology, University of Minnesota, 1999.
- BS Mechanical Engineering, University of Iowa, 1983.

Presentations (last updated Dec. 2016)

International Conference on Web Handling (IWEB)

Tin-Canning Defects in Thin Film Winding, IWEB13, June 2015 Stillwater, OK (2nd Author Kevin Cole, Optimation)

Winding Process Calibration and Comparison, IWEB13, 2015 Stillwater, OK

Wireless Monitoring of Internal Wound Roll Pressures, IWEB12, 2013 Stillwater, OK (2nd Author Robert Updike, Optimation) – Won Best Paper Award

Machine and Transverse Direction Errors in Web Handling and Winding: Root Causes and Solutions, IWEB11, 2013 Stillwater, OK

Wrinkling of Foils, IWEB11, 2011 Stillwater, OK (2nd Author Kevin Cole, Optimation)

Wrinkling of Wide Webs, IWEB11, 2011 Stillwater, OK (2nd Author Kevin Cole, Optimation)

Measurement and Modeling of 2-D and 3-D Web Feed Rates in Rubber Covered Nip Roller Applications, IWEB11, 2011 Stillwater, OK (1st Author Kevin Cole, Optimation)

Taxonomy of Web Wrinkles, IWEB10, 2009 Stillwater, OK

Practical Application and Design of Concave Rollers, IWEB9, 2007 Stillwater, OK

Practical Application of Roller Performance Measurements and Models, IWEB7, 2003 Stillwater, OK

Nip Induced Tension in Draw Control Processes, IWEB3, 1995 Stillwater, OK

Recent Japanese Patents in Web Handling Technology, IWEB1, 1991, Stillwater, OK

Applied/AIMCAL Web Handling Conference (AWEB)

Wireless Monitoring of Winding Roll Pressures, AWEB12, 2010 Myrtle Beach, SC

Wrinkling at Foils, AWEB12, 2010 Myrtle Beach, SC

Wrinkling at Nip Rollers, AWEB10, 2010 Rochester, NY

Wrinkling and Spreading of Wide Webs, AWEB10, 2010 Rochester, NY

Size Matters: The Effects of Roll Diameter on Web Handling, AWEB08, 2008 Minneapolis, MN

Why Isn't Your Slitter Running?, AWEB08, 2008 Minneapolis, MN

Measuring Pressure Variations in Rubber-Steel Nips, AWEB06, 2006 Charlotte, NC

Public Seminars and Workshops

Principles of Web Handling and Winding, Oct 2015, Charlotte NC; Mar 2016, Los Angeles CA; June 2016, Dusseldorf, Germany; Aug 2015, Milwaukee WI

Principles of Web Handing, Mar 2014, Los Angeles CA; May 2015, Manchester, UK; June 2015, Seattle WA; Nov 2015, Munich, Germany

Principles of Winding and Slitting, June 2014, Manchester UK, June 2015, Seattle, WA

Center Winding Workshop, Apr 2011, June 2010, June 2009; Rochester, NY

Wrinkling and Spreading Workshop, Mar 2011, Apr 2010, Nov 2009; Rochester, NY

Nip Roller Systems Workshop, Mar 2011, Feb 2010; Rochester, NY

Roller Performance Workshop, Aug 2010, June 2009; Rochester, NY

General Web Handling Workshop, Dec 2009; Rochester, NY

Seminars For Engineers (S4E)

Web Handling 1: Understanding Web Handling, 2008 Hilton Head, SC; Chicago, IL;

Web Handling 2: Understanding Winding and Slitting, 2008 Charleston, SC; Chicago, IL; Lexington, KY

Oklahoma State University – Web Handling Research Center (OSU WHRC), *Web Handling Applications Seminar* (co-instructor), Fall 1991 to Spring 1995, Stillwater, OK

Association of Industrial Metalizers, Coaters, and Laminators (AIMCAL) / Converting Equipment Manufacturers Association (CEMA)

Why Isn't Your Slitter Running?, AIMCAL Fall Technical Conference, Myrtle Beach, SC, Oct 2005

Reverse Crown and Flat Expander Anti-Wrinkle Rollers – For Better of For Worse, AIMCAL Fall Technical Conference, Charleston, SC, Oct 2004

Winding Techniques, Slitting Techniques, CEMA Slitting and Rewinding Seminar, Philadelphia, PA, Nov 2003; Indianapolis, IN, April 2004; Atlanta, GA, Nov 2004; Chicago, IL, April 2005; Ontario, CA, Nov 2005; New Brunswick, NJ, April 2006; Rosemount, IL, June 2007; Minneapolis, MN, Nov 2007; Charlotte, NC, April 2008; Chicago, IL, Oct 2008.

Roller Performance Measurements and Models, AIMCAL Fall Tech. Conf., Santa Ana Pueblo, NM, Oct 2003

Association of Industrial Metalizers, Coaters, and Laminators (AIMCAL) / Converting Equipment Manufacturers Association (CEMA) [continued]

Laminating Process Fundamentals, CEMA Coating and Laminating Seminar, Charlotte, NC, April 2008; Chicago, IL, June 2007; Chicago, IL, May 2005; May 2003, AIMCAL Fall Tech Conf, Oct 2003

Converting Waste Analysis: A Profit is a Terrible Thing to Waste, 2003 CMM Show, Chicago, IL

Stripe Slitting: Unwind to Rewind Lateral Registration, AIMCAL Fall Technical Conference, Sedona, AZ, Oct 2002

Tension Control for Web Processes, AIMCAL Fall Technical Conference, Hilton Head, SC, Oct 2001

Other Conference Presentations:

ICE USA (International Converting): *Answers to Tension Control Questions*, Orlando, FL April 2013

ICE USA (International Converting): *A Video Guide to Web Wrinkling*, Orlando, FL April 2011

CMM Show (Converting Machinery and Material Show): *Solving Common Tensioning, Slipping, and Scratching Problems*, CMM Show Technical Program (3-hr seminar), Chicago, IL, April 2005, June 2007, June 2009

Converting Technical Institute (CTI): *How to Eliminate Web Handling and Winding Defects in Coating and Converting Processes*, Two-Day Technical Seminar, Tokyo and Osaka, Japan, May 2005, July 2006, May 2009

Technical Association of Pulp and Paper Industry (TAPPI): *Roller Performance Measurements and Models*, TAPPI PLACE, Orlando, FL, Aug 2003; *Winding Tutorial*, TAPPI Spring Technical Conference, Chicago, IL, May 2003; *Web Wrinkles: Diagnoses and Remedies*, TAPPI PLACE (Polymers, Laminates, Adhesives, Coatings, and Extrusion) Technical Conference, Boston, MA, Oct 2002

International Society for Coating Science and Technology (ISCST): *Web Handling Process Fundamentals*, 2001 ISCST Technical Conference, Scottsdale, AZ; *Web Handling Process Fundamentals*, 1999 ISCST Technical Conference, Wilmington, DE

Publications

Paper Film Foil Converter Magazine

2016

Unwind Splice Reliability
Action Plan for Baggy Webs

2015

Air Bubbles in Your Laminate
When Do Small Rollers and Cores Cause Defects?
Unwritten Rules, Now Written Parts 1 and 2
Should Unwind Tension Be Lower Than Winding Tension?
How To Create an MD Running Fold in a Wide Web
Does This Roll Contain Quality Web?

2014

Do Your Homework to Solve Telescoping Roll Problems
All I Ever Needed to Know —I Learned in Elementary School
The Importance of Running Centered
The Case of the Missing Pacer
Globe Trekking Adventures in Web Handling, Winding & Slitting
Wrinkling?
Is There a Doorman at Your Winder or Do You Let Just Any Web Get In?
Do You Need a Roller That Is Chillin' and Groovy?
War on Bagginess
The Case for Shorter (and Longer) Acceleration Times

2013

How the Open Loop Torque Control Sets Winding Tension
What Mechanisms Cause a Web to Shift Laterally?
Curl Mechanics Pt. 2—Transverse Direction Curl
Curl Mechanics Pt. 1—Machine Direction Curl
Eliminating Boundary Layer Air During Winding
Eliminating Tin-Canning Rings on Your Wound Rolls
Winding Process Fundamentals, Pt. 4 – Transverse Direction Variations, Pt. 3 – Roll Strain Changes and Defects, Pt. 2 – Wound-On Tension, Pt. 1 – Winder Design and Tension

2012

To Break or Not To Break--What is the Tension?
Why Bigger Cores?
Why Avoid Nips?
Q&A on Lateral Movement

Paper Film Foil Converter Magazine (cont'd)

2012 (continued)

Measure Tension Wherever Possible
Test Your Web Handling Knowledge
Blocking in Rolls

2011

A Rant from Tim Walker – Please Display Nip and Tension in Engineering Units
Take the Laminator Quiz
Plotting Shear Wrinkles
Wrinkling of Foils
Is Wound Roll Pressure Greater Than Tire Pressure?
Displacement vs Steering | Battle of the Web Guides
Would You Like Nips with That?
Limitations of Vacuum Pull Rollers
Blow Away Your Roller
Don't Flip Out—It's Just a Web Flip
Limitations of Vacuum Pull Rollers
Do You Want Nips with That? Parts 1 & 2

2010

What is a Tension Zone & How Many Are Needed?
Q&A on Roller Deflection Parts 1, 2, and 3
Your Guide to Web Guiding Part I, Part II
Force Is Needed to Shift a Web
Do You Need an Auto Web Guide?
How To Control Roller Nips
Reflections on Deflection
Are You Getting the Shaft

2009

What is the Right Tension?
Cores – The Foundation of Winding
A Torque Devlin in the Details
Time for Unwinding Upgrade?
Winding Doesn't Add Up
Playing the Wrinkle Blame Game
Stress & Strain
The Coefficient of Winding Trouble
The Great Span Length Debate – Part 2
The Great Span Length Debate – Part 1
How Much Misalignment is Trouble – Part 2
How Much Misalignment is Trouble – Part 1

Paper Film Foil Converter Magazine (cont'd)

2008

Who's Driving This Nip?
Shifty Answers to Nip-Induced Tracking,
Do You Have a Need for Speed?
A Wealth of Accumulators
Winding: What We Know & What We Don't Know
Idler Roller Bearings: Living the Good Long Life?
The Spin on Idler Roller Testing
A Recipe for Scratching
Your Guide to the Right Web Guide
Buyers and Suppliers: Can We Dance?
Thinking About New Equipment
Get a Grip: Driving Your Web

2007

Differential Winding Limits: Part I&II
In Search of Tension Isolation
Support Your Rollers
Baggy Webs: Part 1—Nightmares; Part 2 – Measuring; Part 3—Causes; Part 4—Minimizing
Under Pressure (A Brighter Look)
Under Pressure
Don't Get Bent out of Shape
Twist & Shout

2006

How to Drive a Winding Roll,
Dancer Rollers: Trust but verify
Web Tension: A Pop Quiz
Difficult Winding: Part I&II
Whatever Floats Your Web
The Pressure of Winding Rolls
Deflecting Nip Roller Problems?
If Not Rough, How About Groovy?
Are You Rough Enough?
Why Isn't Your Slitter Running
The Secret Strain (feature article)

2005

Converting Rx: Bowed Rollers
The Case for Automatic Splicing
The Converting Relay Race: Parts 1-2
Concave Rollers Pros and Cons
Drawing Conclusions: Parts 1-2
Clean Thinking
Get Out of the Scroll Business
Why Tension?
Who Likes Film?

Paper Film Foil Converter Magazine (cont'd)

2004

The Harms of Harmony
Slitting Debris: Cracking the Case
Ten Tips on Antiwrinkle Rollers
Can This Lamination Be Saved?
The Signs of Shear Wrinkles
Steering Directions Made Simple
Tracking Wrinkles: Parts 1-2
When Rollers Fight, Webs Lose
Fun with Force Gauges
Reduce Web Nip Problems
Friction Circles on A Winter's Day

2003

Linebackers, Leashes, and Superman: A Look at Stress
Small Flexing, Big Benefits
The Long and Short of Web Bending
Where Does a Spreader Spread?
Going with the Parallel Flow
A Slippery Answer to Web Scratching
Winding Better Rolls
Can't Touch This (Web): Parts 1-3
Cinching: Belt Tightening Gone Bad: Parts 1-2

2002

Differential Rewinding: Parts 1-3
Five Questions and Answers About Lubrication
Well Done Tension
Goldilocks, the Three Bears, and Web-Roller Traction
Outsourcing Is Trendy, But Is It Right For Process Expertise?
Stripe Slitting: The Challenge of Staying Within the Lines
Strand Tracking Problems on Your Slitter/Rewinder
Mapping Your Way to Better Tension Control
Pyramids and Wound Rolls: Long-Lasting Quality
Web Line Knowledge Offers A Competitive Advantage

2001 (for Flexible Packaging Magazine

Time to Unwind,
Clean Up Your Act: Five Steps to Clean Manufacturing,
Stretching Exercises
Tale of the Tape

Roll and Web Defect Terminology, 2nd Edition - *Chapter 9 – Web Handling Defects* (covering tension control and slip related defects), TAPPI Press, 2008

TJWA Inc. Consulting Rates and Example Estimate

Consulting Rates

My 2017 consulting rate is $2920 per day, plus travel time and expenses.
Additional time for requested preparation, report, or other non-travel work is $365 per hour.
Training rates are $465 per hour or $3720 per day.
Travel expenses are recharged at costs.
Travel time is charged at $95 per hour (I charge only for flight time or long drives).
Estimate for specific projects will be provided upon request.

Example Estimate

Consulting & Travel Time

$ 2920	= $2920 per consulting day, 1 consulting days requested
$ 730	= $ 365 per hr add'l consulting time, 2 hrs prep or report work
$ 570	= $ 95 per travel time hour, 6 hrs in travel time
$ 4220	total time charges

Travel Estimates

$ 900	airfare
$ 200	rental car
$ 150	hotel
$ 75	taxi to/from airport
$ 55	meal per diem
$ 1370	total travel charges

Total time and travel estimate for 1 day trip is:

$ 4220	time estimate
$ 1370	travel estimate
$ 5590	total estimate:

Additional travel days are estimated at $3240 per day (includes additional time and travel expenses).